**Differentiation rules**

MATH 1334 Wednesday, October 14, 2026

## Our class example: $f(x) = 3x - x^2$

Recall the calculation we did together:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{[3(x+h) - (x+h)^2] - [3x - x^2]}{h} \\ &= \lim_{h\to 0} (3 - 2x - h) = 3 - 2x. \end{aligned}$$

**Question.** Could we obtain $3 - 2x$ directly from the terms $3x$ and $-x^2$? The limit laws let us build a limit from simpler limits. We want rules that let us build a derivative from simpler derivatives.

### Section 3.3: the rules we will use

**Theorem** (Differentiation rules)**.** *Suppose $f$ and $g$ are differentiable at $x$. Let $c$ be a real constant and $n$ any real number. Then*

| | |
|---|---|
| Constant: | $\frac{d}{dx}(c) = 0.$ |
| Identity: | $\frac{d}{dx}(x) = 1.$ |
| Sum: | $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x).$ |
| Difference: | $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x).$ |
| Scalar multiple: | $\frac{d}{dx}[cf(x)] = cf'(x).$ |
| Power: | $\frac{d}{dx}(x^n) = nx^{n-1}.$ |
| Product: | $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$ |
| Quotient: | $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}, \quad g(x) \neq 0.$ |

*The power rule includes negative and fractional exponents. It holds for $x > 0$ for every real $n$, and wherever both sides are defined for other inputs. For $n = 0$, use the constant rule.*

**Compare with the limit laws.** Differentiation distributes over sums and constant multiples. Products and quotients have different formulas: $(fg)'$ is generally not $f'g'$, and $(f/g)'$ is generally not $f'/g'$. The power rule above concerns $x^n$; powers of a function will be treated with the chain rule.

**Our route.** Use and explain the rules; combine them to compute derivatives; then use derivatives with tangent lines.

## Sums, constant multiples, and powers

**Example 1.** Use the differentiation rules to recover the derivative.

$$\begin{aligned}\frac{d}{dx}(3x - x^2) &= 3\frac{d}{dx}(x) - \frac{d}{dx}(x^2) \\ &= 3(1) - 2x = 3 - 2x.\end{aligned}$$

We used the difference, scalar-multiple, identity, and power rules. Why are we allowed to separate the calculation this way?

### Why sums and constant multiples work

For differentiable $u, v$ and a constant $c$,

$$\begin{aligned}\frac{d}{dx}[cu(x) + v(x)] &= \lim_{h\to 0} \frac{c[u(x+h) - u(x)] + [v(x+h) - v(x)]}{h} \\ &= c\lim_{h\to 0} \frac{u(x+h) - u(x)}{h} + \lim_{h\to 0} \frac{v(x+h) - v(x)}{h} \\ &= cu'(x) + v'(x).\end{aligned}$$

The scalar-multiple and sum limit laws justify the second equality. Replacing $v$ by $-v$ gives the difference rule. Also,

$$\frac{d}{dx}(c) = \lim_{h\to 0} \frac{c - c}{h} = 0, \qquad \frac{d}{dx}(x) = \lim_{h\to 0} \frac{x + h - x}{h} = 1.$$

**Example 2.** Differentiate $p(x) = 4x^3 - 2x + 7$.

*Use the same rules.*

$$p'(x) = 4(3x^2) - 2(1) + 0 = 12x^2 - 2.$$

The constant 7 disappears; the derivative of $-2x$ is $-2$.

### Products: one derivative at a time

Our motivating function can also be written $f(x) = x(3 - x)$. Multiplying the derivatives of the two factors would give $1(-1) = -1$, but we already know $f'(x) = 3 - 2x$. That guess fails.

**Example 3.** Apply the product rule to $f(x) = x(3 - x)$:

$$f'(x) = 1(3 - x) + x(-1) = 3 - 2x.$$

Each term differentiates one factor and keeps the other in place.

#### Why are there two terms?

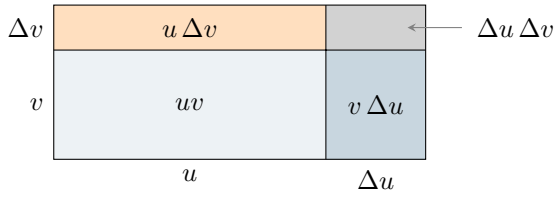
For differentiable factors $u(x)$ and $v(x)$, let

$$\Delta u = u(x + h) - u(x), \qquad \Delta v = v(x + h) - v(x).$$

In the following algebra, $u$ and $v$ denote their values at $x$. Then

$$\Delta(uv) = (u + \Delta u)(v + \Delta v) - uv = v\,\Delta u + u\,\Delta v + \Delta u\,\Delta v.$$

For positive lengths and changes, the three added regions show this change in area. The diagram illustrates the general product rule; it is not a graph of $x(3 - x)$.




The algebra also holds for changes of either sign. Divide by $h$:

$$\frac{\Delta(uv)}{h} = v\frac{\Delta u}{h} + u\frac{\Delta v}{h} + \Delta u\frac{\Delta v}{h} \xrightarrow[h\to 0]{} vu'(x) + uv'(x).$$

The last term tends to zero because differentiability implies continuity: $\Delta u \to 0$, while $\Delta v/h \to v'(x)$ is finite. The left side tends to $(uv)'(x)$, giving the product rule.

**Example 4.** Differentiate $y = (x - 1)(3x^2 + 2)$.

*Product rule.*

$$\begin{aligned} y' &= 1(3x^2 + 2) + (x - 1)(6x) \\ &= 9x^2 - 6x + 2. \end{aligned}$$

Expanding $y = 3x^3 - 3x^2 + 2x - 2$ first gives the same derivative.

## Quotients and negative powers

**Example 5.** Differentiate $y = (x^2+9)/(x^2-1)$.

*Quotient rule.*

$$y' = \frac{(x^2-1)(2x) - (x^2+9)(2x)}{(x^2-1)^2}$$
$$= \frac{2x^3 - 2x - 2x^3 - 18x}{(x^2-1)^2} = \frac{-20x}{(x^2-1)^2}, \qquad x \neq \pm 1.$$

**Example 6.** Differentiate $f(x) = x^4 + 3/x$.

*Rewrite as powers.* For $x \neq 0$,

$$f(x) = x^4 + 3x^{-1} \quad \Longrightarrow \quad f'(x) = 4x^3 - 3x^{-2} = 4x^3 - \frac{3}{x^2}.$$

A fraction may be easier to differentiate after rewriting it. The exponent decreases by one: $-1-1=-2$.

### Combining rules and differentiating again

First look for useful algebra. If a product or quotient remains, identify its two parts and apply the appropriate rule. Simplify the derivative last.

**Example 7.** Differentiate $y = x^2(3/x^2 + 4/x^3)$.

*Simplify first.*

$$y = 3 + \frac{4}{x} \quad \Longrightarrow \quad y' = -\frac{4}{x^2}.$$

Using the product rule gives the same result, with more cancellation:

$$y' = 2x(3x^{-2} + 4x^{-3}) + x^2(-6x^{-3} - 12x^{-4}) = -4x^{-2}.$$

**Example 8.** Differentiate $y = 2x\left(5x^3 + \dfrac{7}{x+2}\right)$.

*A product containing a quotient.* First find the derivative inside:

$$\frac{d}{dx}\left(\frac{7}{x+2}\right) = \frac{(x+2)(0) - 7(1)}{(x+2)^2} = -\frac{7}{(x+2)^2}.$$

Now apply the product rule to the entire expression:

$$\begin{aligned} y' &= 2\left(5x^3 + \frac{7}{x+2}\right) + 2x\left(15x^2 - \frac{7}{(x+2)^2}\right) \\ &= 40x^3 + \frac{14(x+2) - 14x}{(x+2)^2} = 40x^3 + \frac{28}{(x+2)^2}, \qquad x \neq -2. \end{aligned}$$

Write a correct derivative before trying to simplify it.

### A second derivative is a derivative of a derivative

If $y = f(x)$, then $y'' = f''(x) = \dfrac{d}{dx}[f'(x)]$, when this derivative exists. Differentiate the first derivative; do not square it.

**Example 9.** Differentiate $y = (3x^3 - x + 2)/x^2$ twice.

*Divide each term by the denominator.*

$$\begin{aligned} y &= 3x - x^{-1} + 2x^{-2}, \\ y' &= 3 + x^{-2} - 4x^{-3}, \\ y'' &= -2x^{-3} + 12x^{-4}, \qquad x \neq 0. \end{aligned}$$

### Tangent lines and prescribed slopes

Once $f'(x)$ is known, a tangent problem becomes a point-and-slope problem:

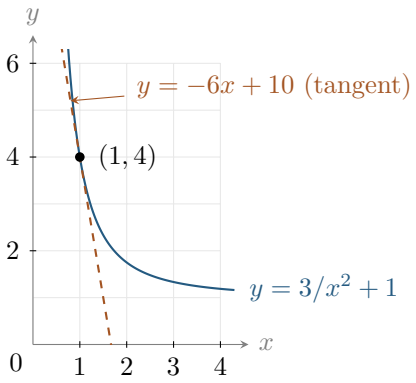
$$y - f(a) = f'(a)(x - a).$$

**Example 10.** Find the tangent to $f(x) = 3/x^2 + 1$ at $x = 1$.

*Point, slope, line.*

$$f(1) = 4, \qquad f'(x) = -6x^{-3}, \qquad f'(1) = -6.$$

Hence $y - 4 = -6(x - 1)$, or $y = -6x + 10$. The graph shows the positive branch and its tangent. At $(1, 4)$ they have both the same value and the same slope; sharing a point alone is not enough.




**Example 11.** For $f(x) = x^3 - 3x^2 + 2$, find all points where the tangent (a) is horizontal and (b) has slope 9.

*Solve for the input, then find the point.* First,

$$f'(x) = 3x^2 - 6x.$$

**(a)** A horizontal tangent has slope 0:

$$3x(x - 2) = 0 \quad \Longrightarrow \quad x = 0, 2.$$

Since $f(0) = 2$ and $f(2) = -2$, the points are $(0, 2)$ and $(2, -2)$.

**(b)** Set $f'(x) = 9$, not $f(x) = 9$:

$$3x^2 - 6x = 9 \quad \Longrightarrow \quad (x - 3)(x + 1) = 0.$$

The inputs are $x = -1, 3$. Evaluating the original function gives the points $(-1, -2)$ and $(3, 2)$. A slope condition specifies $f'$, while coordinates require $f$.

### Section 3.3 recap

Choose the form, apply the rule, then simplify. For tangent lines, find both the point and the slope.